

PURE BAKE COOKIES SALES ANALYSIS

1. Project Overview

The **Pure Bake Sales Analysis Dashboard** is developed using Excel for CLEANING, AGGREGATION process and Power BI to provide a comprehensive and interactive DASHBOARD view of sales performance for a COOKIES business. This project focuses on transforming raw sales data into meaningful insights through visually appealing dashboards and reports.

Objective

The dashboard enables stakeholders to monitor key business metrics such as total sales performance, product-wise performance and revenue growth. By leveraging data visualization tools like charts, filters, the system helps in identifying patterns, seasonal trends, and high-performing products.

2. Data Sources

- **Source Description and Timeline:** Google Dataset Search - 2020-2025.
- **Domain:** Sales Analytics

1. Problem Statement

- To analyses total quantity of cookies sales, total revenue, total profit and profit percentage.
 - To analyse month wise performance.
 - To analyse sales performance across regions and products to identify top-performing areas
 - To analyse profit trends of week type and sales quantity by week type.
 - To evaluate employee performance of top one.
 - To understand customer purchasing behaviour to enhance marketing and retention strategies.
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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Transaction Date	Date	The date on which the purchase transaction occurred
Buyer Name	String (Text)	Name of the customer who made the purchase
Buyer Location	Text (Categorical)	Geographic location (city/area) of the buyer
Buyer date of birth	Date	Customer's birth date used to calculate age
Payment Mode	Text (Categorical)	Method of payment (Cash, Card, UPI, etc.)
Quantity Purchased	Numeric (Integer)	Number of items bought in a transaction
Product Code	Text (Identifier)	Unique code assigned to each product
Sales Representative	Text (Categorical)	Name of the employee handling the sale
Gender	Text (Categorical)	Gender of the customer (Male/Female/Other)
Discount	Numeric (Decimal)	Discount applied on the purchase (in % or amount)
Biscuit Brand	Text (Categorical)	Brand name of the biscuit product sold
Weekday	Text (Derived)	Day of the week (Monday–Sunday) from transaction date
Week Type	Text (Derived)	Classification as Weekday or Weekend
Age-Group	Text (Derived)	Customer category based on age (e.g., 18–25, 26–35, etc.)

5. Tools & Technologies

- **Excel:** Data cleaning and transformation.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Handled missing values, standardized date formats, Proper case and trim the spaces, removed un wanted rows and created calculated fields.

Transaction Date format:

Transaction Date	Transaction Date
45342	01/01/2024
45366	01/01/2024
45304	02/01/2024
45421	02/01/2024
45432	02/01/2024
45368	02/01/2024
45339	02/01/2024
45441	02/01/2024
45410	02/01/2024
45329	03/01/2024
45473	03/01/2024
45337	03/01/2024
45321	03/01/2024
45377	04/01/2024
45500	04/01/2024
45465	04/01/2024
45356	04/01/2024
45494	04/01/2024
45473	04/01/2024
45492	05/01/2024
45339	05/01/2024
45391	05/01/2024
45419	05/01/2024
45432	05/01/2024
45293	05/01/2024
45455	05/01/2024

Buyer name concatenate format:

Buyer First Name	Buyer Last Name
Allen	Santos
Brenda	Shaw
Robert	Hernandez
Alexander	Smith
Mary	Rodriguez
Clinton	Haynes
Jamie	Steele
DAVID	Hunt
Robert	Hernandez
Alexander	Smith
Mary	Rodriguez
Samantha	Wong
Gregory	Gibson
Robert	Hernandez
Alexander	Smith
Mary	Rodriguez
SARAH	Case
Taylor	Wilson
Robert	Hernandez
Alexander	Smith
Mary	Rodriguez

Transaction Date	Buyer Name	Buyer First Name	Buyer Last Name	Buyer Location	Buyer Date of Birth	Payment Method	Quantity Purch.	Product Code	Buyer Gender	Buyer Status
1/1/2024	William Manning	William	Manning	Phoenix	2/6/1980 0:00	Debit Card	134	Prod07	Sara Robet	Male
1/1/2024	Karen Duke	Karen	Duke	Dallas	2/20/1978 0:00	Cash	454	Prod04	Susan Dear	Male
1/2/2024	Alexander Smith	Alexander	Smith	San Jose	9/26/1987 0:00	Cash	428	Prod04	Susan Dear	Male
1/2/2024	REBECCA Larson	REBECCA	Larson	Unknown	1/25/1961 0:00	Credit Card	209		Katie Stant	Other
1/2/2024	Antonio Miller	Antonio	Miller	Phoenix	3/20/1996 0:00	Cash	212	Prod04	Sara Robet	Male
1/2/2024	Christian Chambers	Christian	Chambers	Philadelphia	11/30/1955 0:00	Debit Card	377	Prod09	April Watki	Male
1/2/2024	Miguel Carlson	Miguel	Carlson	San Antonio	4/6/1969 0:00	Credit Card	305	Prod06	Katie Stant	Male
1/2/2024	James Ryan	James	Ryan	San Antonio	4/6/1973 0:00	Mobile Payment	263	Prod09	Katie Stant	Male
1/2/2024	Brianna Bender	Brianna	Bender	San Antonio	10/23/1975 0:00	Credit Card	434	Prod05	Susan Dear	Male
1/3/2024	Lawrence Cruz	Lawrence	Cruz	Philadelphia	4/28/1972 0:00	Mobile Payment	75	Prod08	Susan Dear	Male
1/3/2024	Daniel Davenport	Daniel	Davenport	Dallas	1/16/1952 0:00	Cash	358		Katie Stant	Male
1/3/2024	Kevin Santiago	Kevin	Santiago	San Jose	8/21/1960 0:00	Mobile Payment	379	Prod09	Susan Dear	Male
1/3/2024	Gregory King	Gregory	King	San Antonio	12/6/1948 0:00	Mobile Payment	347	Prod05	Sara Robet	Male
1/4/2024	George Stevens	George	Stevens	Unknown	11/29/1991 0:00	Credit Card	338	Prod10	Susan Dear	Male
1/4/2024	Timothy Roberts	Timothy	Roberts	Unknown	11/26/1948 0:00	Debit Card	46	Prod04	Susan Dear	Male
1/4/2024	Heidi Doyle	Heidi	Doyle	San Diego	1/10/1981 0:00	Credit Card	26	Prod07	Katie Stant	Male
1/4/2024	Randall Best	Randall	Best	San Jose	4/25/1945 0:00	Cash	199	Prod05	April Watki	Male
1/4/2024	Bradley Rosales	Bradley	Rosales	Los Angeles	7/29/1991 0:00	Mobile Payment	63	Prod02	Sara Robet	Male
1/4/2024	Mary Lopez	Mary	Lopez	Chicago	9/4/1964 0:00	Cash	413	Prod10	Susan Dear	Female
1/4/2024	Timothy Perez	Timothy	Perez	San Jose	9/24/2004 0:00	Cash	175	Prod04	Susan Dear	Male
1/5/2024	Mark Williams	Mark	Williams	Unknown	9/19/1985 0:00	Debit Card	224	Prod04	Katie Stant	Male
1/5/2024	David Barr	David	Barr	New York	1/29/1984 0:00	Debit Card	104	Prod01	Susan Dear	Other
1/5/2024	Frank Lee	Frank	Lee	San Diego	12/20/1947 0:00	Credit Card	82	Prod08	Sara Robet	Male
1/5/2024	Nicole Thompson	Nicole	Thompson	Phoenix	9/25/1980 0:00	Credit Card	277	Prod04	Sara Robet	Male
1/5/2024	Scott Watson	Scott	Watson	New York	11/3/1982 0:00	Credit Card	180	Prod10	Susan Dear	Male
1/5/2024	Cindy Barrett	Cindy	Barrett	San Jose	8/5/2005 0:00	Debit Card	108	Prod03	Sara Robet	Male

Proper case buyer's name:

Buyer Name	Buyer Name2
Hannah Jones	Hannah Jones
Kyle Henderson	Kyle Henderson
Marcia Johnson	Marcia Johnson
Thomas Carter	THOMAS Carter
Carol Young	CAROL Young
Jesse Moreno	Jesse Moreno
Barbara Nelson	Barbara Nelson
Alexander Smith	Alexander Smith
Melissa Curtis	Melissa Curtis
Timothy Oconnor	Timothy Oconnor
Joseph Glover	Joseph Glover
Samantha Marks	Samantha Marks
Jillian Burnett	Jillian Burnett
Cole Curry	Cole Curry
Ashley Sanford	Ashley Sanford
Frank Fletcher	Frank Fletcher

Removing duplicates

Transaction ID	Buyer Name	Buyer Location	Buyer Date of Birth	Payment Method	Quantity Purch	Product Code	Representative	Gender	Discount
02/01/2024	William Manning	Phoenix	2/6/1980 0:00	Debit Card	134	Pro07	Sara Robert	Male	0.1
02/01/2024	Karen Duke	Dallas	2/20/1978 0:00	Cash	454	Pro04	Susan Dear	Male	0.1
02/01/2024	Alexander Smith	San Jose	9/26/1987 0:00	Cash	428	Pro04	Susan Dear	Male	0.1
02/01/2024	REBECCA Larson	Unknown	1/25/1981 0:00	Credit			Katie Stant	Other	0.1
02/01/2024	Antonio Miller	Phoenix	3/20/1996 0:00	Cash			Sara Robert	Male	0.1
02/01/2024	Christian Chambers	Philadelphia	11/30/1955 0:00	Debit			April Watki	Male	0.1
02/01/2024	Miguel Carlson	San Antonio	4/6/1980 0:00	Credit			Katie Stant	Male	0.1
02/01/2024	James Ryan	San Antonio	4/6/1973 0:00	Mobile			Katie Stant	Male	0.1
02/01/2024	Brianna Bender	San Antonio	10/23/1975 0:00	Credit			Susan Dear	Male	0.1
02/01/2024	Lawrence Cruz	Philadelphia	4/28/1972 0:00	Mobile			Susan Dear	Male	0.1
02/01/2024	Daniel Davenport	Dallas	1/16/1963 0:00	Cash			Katie Stant	Male	0.1
02/01/2024	Kevin Santiago	San Jose	8/21/1960 0:00	Mobile			Susan Dear	Male	0.1
02/01/2024	Gregory King	San Antonio	12/6/1949 0:00	Mobile Payment	263	Pro09	Sara Robert	Male	0.1
02/01/2024	George Stevens	Unknown	11/26/1993 0:00	Credit Card	338	Pro10	Susan Dear	Male	0.1
02/01/2024	Timothy Roberts		11/26/1949 0:00	Debit Card	46	Pro04	Susan Dear	Male	0.1
02/01/2024	Held Doyle	San Diego	1/10/1981 0:00	Credit Card	26	Pro07	Katie Stant	Male	0.1
02/01/2024	Randall Best	San Jose	4/25/1945 0:00	Cash	199	Pro05	April Watki	Male	0.1
02/01/2024	Bradley Rosales	Los Angeles	7/29/1991 0:00	Mobile Payment	63	Pro02	Sara Robert	Male	0.1
02/01/2024	Mary Lopez	Chicago	9/4/1964 0:00	Cash	413	Pro10	Susan Dear	Female	0.1
02/01/2024	Timothy Perez	San Jose	9/24/2004 0:00	Cash	175	Pro04	Susan Dear	Male	0.1
02/01/2024	Mark Williams	Unknown	9/19/1985 0:00	Debit Card	224	Pro04	Katie Stant	Male	0.1
02/01/2024	David Barr	New York	5/29/1984 0:00	Debit Card	104	Pro02	Susan Dear	Other	0.1
02/01/2024	Frank Lee	San Diego	12/20/1947 0:00	Credit Card	82	Pro08	Sara Robert	Male	0.1
02/01/2024	Nicole Thompson	Phoenix	9/25/1980 0:00	Credit Card	277	Pro04	Sara Robert	Male	0.1
02/01/2024	Scott Watson	New York	11/3/1982 0:00	Credit Card	180	Pro10	Susan Dear	Male	0.1
02/01/2024	Cindy Barrett	San Jose	8/5/2005 0:00	Debit Card	108	Pro03	Sara Robert	Male	0.1

Gender missing values:

Sales Representative	Gender	Disc
Sort A to Z		
Sort Z to A		
Sort by Color		
Sheet View		
Clear Filter From "Gender"		
Filter by Color		
Text Filters		
Search		
(Select All)		
☒ Female		
☒ Male		
☒ (Blanks)		
OK		
Cancel		

Payment Meth	Quantity Purch	Product Code	Sales Representative	Gender	Gender	Discou			
Debit Card	134	Pro07	Sara Robertson	Male	Male	0.1			
Cash	454	Pro04	Susan Dean	Male	Male	0.1			
Cash	428	Pro04	Susan Dean	Male	Male	0.1			261
Credit Card	209	Unknown	Katie Stanton	Other	Other	0.1			
Cash	212	Pro04	Sara Robertson	Male	Male	0.1			
Debit Card	377	Pro09	April Watkins	Male	Male	0.1			
Credit Card	305	Pro06	Katie Stanton	Male	Male	0.1			
Mobile Payment	263	Pro09	Katie Stanton	Male	Male	0.1			
Credit Card	434	Pro05	Susan Dean	Male	Male	0.1			
Mobile Payment	75	Pro08	Susan Dean	Male	Male	0.1			
Cash	358	Unknown	Katie Stanton	Male	Male	0.1			
Mobile Payment	379	Pro09	Susan Dean	Male	Male	0.1			
Mobile Payment	347	Pro05	Sara Robertson	Male	Male	0.1			
Credit Card	338	Pro10	Susan Dean	Male	Male	0.1			
Debit Card	46	Pro04	Susan Dean	Male	Male	0.1			

Payment mode missing values fill:

Buyer Date of Birth ▾ | Payment Method ▾ | Qu

A Z Sort A to Z

Z A Sort Z to A

Sort by Color >

Sheet View >

Clear Filter From "Payment Method"

Filter by Color >

Text Filters >

Search

- ☒ (Select All)
- ☒ Cash
- ☒ Credit Card
- ☒ Debit Card
- ☒ Mobile Payment
- ☒ (Blanks)

OK Cancel

=IF(ISBLANK(F2), INDEX(\$F\$2:\$F\$1000, MATCH(MAX(COUNTIF(\$F\$2:\$F\$1000, \$F\$2:\$F\$1000)), COUNTIF(\$F\$2:\$F\$1000, \$F\$2:\$F\$1000), 0)), F2)								
Er Name	Buyer Location	Buyer Date of Birth	Payment Method	Payment Method2	Quantity Purch	Product Code	Representative	Gender
Manning	Phoenix	2/6/1980	Debit Card	Debit Card	134	Pro07	Sara Rober	Male
Jke	Dallas	2/20/1978	Cash	Cash	454	Pro04	Susan Dear	Male
er Smith	San Jose	9/26/1987	Cash	Cash	428	Pro04	Susan Dear	Male
CA Larson	Unknown	1/25/1961	Credit Card	Credit Card	209		Katie Stant	Other
Miller	Phoenix	3/20/1996	Cash	Cash	212	Pro04	Sara Rober	Male
Chambers	Philadelphia	11/30/1955	Debit Card	Debit Card	377	Pro09	April Watki	Male
arlson	San Antonio	4/6/1969	Credit Card	Credit Card	305	Pro06	Katie Stant	Male
yan	San Antonio	4/6/1973	Mobile Payment	Mobile Payment	263	Pro09	Katie Stant	Male
Bender	San Antonio	10/23/1975	Credit Card	Credit Card	434	Pro05	Susan Dear	Male
e Cruz	Philadelphia	4/28/1972	Mobile Payment	Mobile Payment	75	Pro08	Susan Dear	Male
avenport	Dallas	1/16/1952	Cash	Cash	358		Katie Stant	Male
ntiago	San Jose	8/21/1960	Mobile Payment	Mobile Payment	379	Pro09	Susan Dear	Male
King	San Antonio	12/6/1949	Mobile Payment	Mobile Payment	347	Pro05	Sara Rober	Male
Stevens	Unknown	11/26/1993	Credit Card	Credit Card	338	Pro10	Susan Dear	Male
Roberts	Unknown	11/16/1949	Debit Card	Debit Card	46	Pro04	Susan Dear	Male
yle	San Diego	1/10/1981	Credit Card	Credit Card	26	Pro07	Katie Stant	Male

Quantity missing values fill:

The screenshot shows the 'Filter by Color' dialog box. The 'Filter by Color' dropdown is set to 'Color'. The 'Search' box is empty. The list of colors includes 491, 492, 493, 494, 495, 496, 497, 500, and (Blanks), all of which are checked. The 'OK' and 'Cancel' buttons are at the bottom.

[illegible]

X Lookups for cost:

=XLOOKUP([@[Product Code]],Product_Code,Cost,0)										
Quantity Purchased	Product Code	Sales Representative	Gender	Discount	Biscuit Brand	Cost				
34	Pro07	Sara Robertson	Male	10%	Oatmeal Cookies	19				
54	Pro04	Susan Dean	Male	10%	Cinnamon Swirls	10				
28	Pro04	Susan Dean	Male	10%	Cinnamon Swirls	10				
09	Unknown	Katie Stanton	Other	10%	Unknown	0				
12	Pro04	Sara Robertson	Male	10%	Cinnamon Swirls	10				

X Lookups for biscuit brand:

=XLOOKUP([@[Product Code]],Product_Code,Biscuit_Brand,"Unknown")						
F	G	H	I	J	K	L
Quantity Purchased	Product Code	Sales Representative	Gender	Discount	Biscuit Brand	
1	Pro07	Sara Robertson	Male	10%	Oatmeal Cookies	
2	Pro04	Susan Dean	Male	10%	Cinnamon Swirls	
3	Pro04	Susan Dean	Male	10%	Cinnamon Swirls	
4	Unknown	Katie Stanton	Other	10%	Unknown	
5	Pro04	Sara Robertson	Male	10%	Cinnamon Swirls	
6	Pro09	April Watkins	Male	10%	Shortbread	
7	Pro06	Katie Stanton	Male	10%	Lemon Crisps	
8	Pro09	Katie Stanton	Male	10%	Shortbread	
9	Pro05	Susan Dean	Male	10%	Ginger Snaps	

COGS:

=[@Quantity Purchased]]*[@Cost]							
J	K	L	M	N	O		
Count	Biscuit Brand	Cost	UnitPrice	Age	COGS		Revenue
10%	Oatmeal Cookies	₹ 19	₹ 23	43	₹ 2,546		
10%	Cinnamon Swirls	₹ 10	₹ 14	45	₹ 4,540		
10%	Cinnamon Swirls	₹ 10	₹ 14	36	₹ 4,280		
10%	Unknown	₹ 0	₹ 0	62	₹ 0		
10%	Cinnamon Swirls	₹ 10	₹ 14	27	₹ 2,120		
10%	Shortbread	₹ 10	₹ 45	68	₹ 3,770		
10%	Lemon Crisps	₹ 7	₹ 9	54	₹ 2,135		
10%	Shortbread	₹ 10	₹ 45	50	₹ 2,630		
10%	Ginger Snaps	₹ 14	₹ 28	48	₹ 6,076		
10%	Peanut Butter Bites	₹ 5	₹ 7	51	₹ 375		
10%	Unknown	₹ 0	₹ 0	71	₹ 0		
10%	Shortbread	₹ 10	₹ 45	63	₹ 3,790		
10%	Ginger Snaps	₹ 14	₹ 28	74	₹ 4,858		
10%	Vanilla Wafers	₹ 18	₹ 36	30	₹ 6,084		
10%	Cinnamon Swirls	₹ 10	₹ 14	74	₹ 460		
10%	Oatmeal Cookies	₹ 19	₹ 23	42	₹ 494		
10%	Ginger Snaps	₹ 14	₹ 28	78	₹ 2,786		

Revenue:

=[@Quantity Purchased]]*[@UnitPrice]*(1-[@Discount])							
J	K	L	M	N	O	P	
Count	Biscuit Brand	Cost	UnitPrice	Age	COGS	Revenue	Profit
10%	Oatmeal Cookies	₹ 19	₹ 23	43	₹ 2,546	₹ 2,774	
10%	Cinnamon Swirls	₹ 10	₹ 14	45	₹ 4,540	₹ 5,720	
10%	Cinnamon Swirls	₹ 10	₹ 14	36	₹ 4,280	₹ 5,393	
10%	Unknown	₹ 0	₹ 0	62	₹ 0	₹ 0	
10%	Cinnamon Swirls	₹ 10	₹ 14	27	₹ 2,120	₹ 2,671	
10%	Shortbread	₹ 10	₹ 45	68	₹ 3,770	₹ 15,269	
10%	Lemon Crisps	₹ 7	₹ 9	54	₹ 2,135	₹ 2,471	
10%	Shortbread	₹ 10	₹ 45	50	₹ 2,630	₹ 10,652	
10%	Ginger Snaps	₹ 14	₹ 28	48	₹ 6,076	₹ 10,937	
10%	Peanut Butter Bites	₹ 5	₹ 7	51	₹ 375	₹ 473	
10%	Unknown	₹ 0	₹ 0	71	₹ 0	₹ 0	
10%	Shortbread	₹ 10	₹ 45	63	₹ 3,790	₹ 15,350	
10%	Ginger Snaps	₹ 14	₹ 28	74	₹ 4,858	₹ 8,744	
10%	Vanilla Wafers	₹ 18	₹ 36	30	₹ 6,084	₹ 10,951	
10%	Cinnamon Swirls	₹ 10	₹ 14	74	₹ 460	₹ 580	

Profit:

=[@Revenue]-[@COGS]							
J	K	L	M	N	O	P	Q
Count	Biscuit Brand	Cost	UnitPrice	Age	COGS	Revenue	Profit
10%	Oatmeal Cookies	₹ 19	₹ 23	43	₹ 2,546	₹ 2,774	₹ 228
10%	Cinnamon Swirls	₹ 10	₹ 14	45	₹ 4,540	₹ 5,720	₹ 1,180
10%	Cinnamon Swirls	₹ 10	₹ 14	36	₹ 4,280	₹ 5,393	₹ 1,113
10%	Unknown	₹ 0	₹ 0	62	₹ 0	₹ 0	₹ 0
10%	Cinnamon Swirls	₹ 10	₹ 14	27	₹ 2,120	₹ 2,671	₹ 551
10%	Shortbread	₹ 10	₹ 45	68	₹ 3,770	₹ 15,269	₹ 11,499
10%	Lemon Crisps	₹ 7	₹ 9	54	₹ 2,135	₹ 2,471	₹ 336
10%	Shortbread	₹ 10	₹ 45	50	₹ 2,630	₹ 10,652	₹ 8,022
10%	Ginger Snaps	₹ 14	₹ 28	48	₹ 6,076	₹ 10,937	₹ 4,861
10%	Peanut Butter Bites	₹ 5	₹ 7	51	₹ 375	₹ 473	₹ 98
10%	Unknown	₹ 0	₹ 0	71	₹ 0	₹ 0	₹ 0
10%	Shortbread	₹ 10	₹ 45	63	₹ 3,790	₹ 15,350	₹ 11,560
10%	Ginger Snaps	₹ 14	₹ 28	74	₹ 4,858	₹ 8,744	₹ 3,886
10%	Vanilla Wafers	₹ 18	₹ 36	30	₹ 6,084	₹ 10,951	₹ 4,867

Weekday:

=WEEKDAY([@[Transaction Date]])									
J	K	L	M	N	O	P	Q	R	
count	Biscuit Brand	Cost	UnitPrice	Age	COGS	Revenue	Profit	Weekday	
10%	Oatmeal Cookies	₹ 19	₹ 23	43	₹ 2,546	₹ 2,774	₹ 228	Monday	
10%	Cinnamon Swirls	₹ 10	₹ 14	45	₹ 4,540	₹ 5,720	₹ 1,180	Monday	
10%	Cinnamon Swirls	₹ 10	₹ 14	36	₹ 4,280	₹ 5,393	₹ 1,113	Tuesday	
10%	Unknown	₹ 0	₹ 0	62	₹ 0	₹ 0	₹ 0	Tuesday	
10%	Cinnamon Swirls	₹ 10	₹ 14	27	₹ 2,120	₹ 2,671	₹ 551	Tuesday	
10%	Shortbread	₹ 10	₹ 45	68	₹ 3,770	₹ 15,269	₹ 11,499	Tuesday	
10%	Lemon Crisps	₹ 7	₹ 9	54	₹ 2,135	₹ 2,471	₹ 336	Tuesday	
10%	Shortbread	₹ 10	₹ 45	50	₹ 2,630	₹ 10,652	₹ 8,022	Tuesday	
10%	Ginger Snaps	₹ 14	₹ 28	48	₹ 6,076	₹ 10,937	₹ 4,861	Tuesday	
10%	Peanut Butter Bites	₹ 5	₹ 7	51	₹ 375	₹ 473	₹ 98	Wednesday	
10%	Unknown	₹ 0	₹ 0	71	₹ 0	₹ 0	₹ 0	Wednesday	
10%	Shortbread	₹ 10	₹ 45	63	₹ 3,790	₹ 15,350	₹ 11,560	Wednesday	
10%	Ginger Snaps	₹ 14	₹ 28	74	₹ 4,858	₹ 8,744	₹ 3,886	Wednesday	
10%	Vanilla Wafers	₹ 18	₹ 36	30	₹ 6,084	₹ 10,951	₹ 4,867	Thursday	
10%	Cinnamon Swirls	₹ 10	₹ 14	74	₹ 460	₹ 580	₹ 120	Thursday	
10%	Oatmeal Cookies	₹ 19	₹ 23	42	₹ 494	₹ 538	₹ 44	Thursday	
10%	Ginger Snaps	₹ 14	₹ 28	78	₹ 2,786	₹ 5,015	₹ 2,229	Thursday	

Week Type:

=IF(WEEKDAY([@[Transaction Date]],2)<=5,"Weekday","Weekend")									
J	K	L	M	N	O	P	Q	R	S
ount	Biscuit Brand	Cost	UnitPrice	Age	COGS	Revenue	Profit	Weekday	Week Type
10%	Oatmeal Cookies	₹ 19	₹ 23	43	₹ 2,546	₹ 2,774	₹ 228	Monday	Weekday
10%	Cinnamon Swirls	₹ 10	₹ 14	45	₹ 4,540	₹ 5,720	₹ 1,180	Monday	Weekday
10%	Cinnamon Swirls	₹ 10	₹ 14	36	₹ 4,280	₹ 5,393	₹ 1,113	Tuesday	Weekday
10%	Unknown	₹ 0	₹ 0	62	₹ 0	₹ 0	₹ 0	Tuesday	Weekday
10%	Cinnamon Swirls	₹ 10	₹ 14	27	₹ 2,120	₹ 2,671	₹ 551	Tuesday	Weekday
10%	Shortbread	₹ 10	₹ 45	68	₹ 3,770	₹ 15,269	₹ 11,499	Tuesday	Weekday
10%	Lemon Crisps	₹ 7	₹ 9	54	₹ 2,135	₹ 2,471	₹ 336	Tuesday	Weekday
10%	Shortbread	₹ 10	₹ 45	50	₹ 2,630	₹ 10,652	₹ 8,022	Tuesday	Weekday
10%	Ginger Snaps	₹ 14	₹ 28	48	₹ 6,076	₹ 10,937	₹ 4,861	Tuesday	Weekday
10%	Peanut Butter Bites	₹ 5	₹ 7	51	₹ 375	₹ 473	₹ 98	Wednesday	Weekday
10%	Unknown	₹ 0	₹ 0	71	₹ 0	₹ 0	₹ 0	Wednesday	Weekday
10%	Shortbread	₹ 10	₹ 45	63	₹ 3,790	₹ 15,350	₹ 11,560	Wednesday	Weekday
10%	Ginger Snaps	₹ 14	₹ 28	74	₹ 4,858	₹ 8,744	₹ 3,886	Wednesday	Weekday
10%	Vanilla Wafers	₹ 18	₹ 36	30	₹ 6,084	₹ 10,951	₹ 4,867	Thursday	Weekday
10%	Cinnamon Swirls	₹ 10	₹ 14	74	₹ 460	₹ 580	₹ 120	Thursday	Weekday
10%	Oatmeal Cookies	₹ 19	₹ 23	42	₹ 494	₹ 538	₹ 44	Thursday	Weekday
10%	Ginger Snaps	₹ 14	₹ 28	78	₹ 2,786	₹ 5,015	₹ 2,229	Thursday	Weekday
10%	Caramel Biscuits	₹ 18	₹ 22	32	₹ 1,134	₹ 1,247	₹ 113	Thursday	Weekday

Age-Group:

=FLOOR([@Age],15)&"-"&FLOOR([@Age],15)+14										
	K	L	M	N	O	P	Q	R	S	T
	Biscuit Brand	Cost	UnitPrice	Age	COGS	Revenue	Profit	Weekday	Week Type	Age-Gr
%	Oatmeal Cookies	₹ 19	₹ 23	43	₹ 2,546	₹ 2,774	₹ 228	Monday	Weekday	30-44
%	Cinnamon Swirls	₹ 10	₹ 14	45	₹ 4,540	₹ 5,720	₹ 1,180	Monday	Weekday	45-59
%	Cinnamon Swirls	₹ 10	₹ 14	36	₹ 4,280	₹ 5,393	₹ 1,113	Tuesday	Weekday	30-44
%	Unknown	₹ 0	₹ 0	62	₹ 0	₹ 0	₹ 0	Tuesday	Weekday	60-74
%	Cinnamon Swirls	₹ 10	₹ 14	27	₹ 2,120	₹ 2,671	₹ 551	Tuesday	Weekday	15-29
%	Shortbread	₹ 10	₹ 45	68	₹ 3,770	₹ 15,269	₹ 11,499	Tuesday	Weekday	60-74
%	Lemon Crisps	₹ 7	₹ 9	54	₹ 2,135	₹ 2,471	₹ 336	Tuesday	Weekday	45-59
%	Shortbread	₹ 10	₹ 45	50	₹ 2,630	₹ 10,652	₹ 8,022	Tuesday	Weekday	45-59
%	Ginger Snaps	₹ 14	₹ 28	48	₹ 6,076	₹ 10,937	₹ 4,861	Tuesday	Weekday	45-59
%	Peanut Butter Bites	₹ 5	₹ 7	51	₹ 375	₹ 473	₹ 98	Wednesday	Weekday	45-59
%	Unknown	₹ 0	₹ 0	71	₹ 0	₹ 0	₹ 0	Wednesday	Weekday	60-74
%	Shortbread	₹ 10	₹ 45	63	₹ 3,790	₹ 15,350	₹ 11,560	Wednesday	Weekday	60-74
%	Ginger Snaps	₹ 14	₹ 28	74	₹ 4,858	₹ 8,744	₹ 3,886	Wednesday	Weekday	60-74
%	Vanilla Wafers	₹ 18	₹ 36	30	₹ 6,084	₹ 10,951	₹ 4,867	Thursday	Weekday	30-44
%	Cinnamon Swirls	₹ 10	₹ 14	74	₹ 460	₹ 580	₹ 120	Thursday	Weekday	60-74
%	Oatmeal Cookies	₹ 19	₹ 23	42	₹ 494	₹ 538	₹ 44	Thursday	Weekday	30-44
%	Ginger Snaps	₹ 14	₹ 28	78	₹ 2,786	₹ 5,015	₹ 2,229	Thursday	Weekday	75-89
%	Caramel Biscuits	₹ 18	₹ 22	32	₹ 1,134	₹ 1,247	₹ 113	Thursday	Weekday	30-44
%	Vanilla Wafers	₹ 18	₹ 36	59	₹ 7,434	₹ 13,381	₹ 5,947	Thursday	Weekday	45-59
%	Cinnamon Swirls	₹ 10	₹ 14	19	₹ 1,750	₹ 2,205	₹ 455	Thursday	Weekday	15-29
%	Cinnamon Swirls	₹ 10	₹ 14	38	₹ 2,240	₹ 2,822	₹ 582	Friday	Weekday	30-44
%	Butter Crunch	₹ 10	₹ 11	39	₹ 1,040	₹ 1,030	₹ 10	Friday	Weekday	30-44

- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

KPIs				
Quantity Purchased	COGS	Revenue	Profit	% Profit Margin
257,159	₹ 2,996,089	₹ 4,824,508	₹ 1,828,419	37.9%

Profitability View	
Biscuit Brand	Profit
Shortbread	₹ 821,701
Buyer Location	Profit
Houston	₹ 214,344
Buyer Name	Profit
Alexander Smith	₹ 42,969
Sales Represent	Profit
Susan Dean	₹ 656,184
Age-Group	Profit
60-74	₹ 444,201

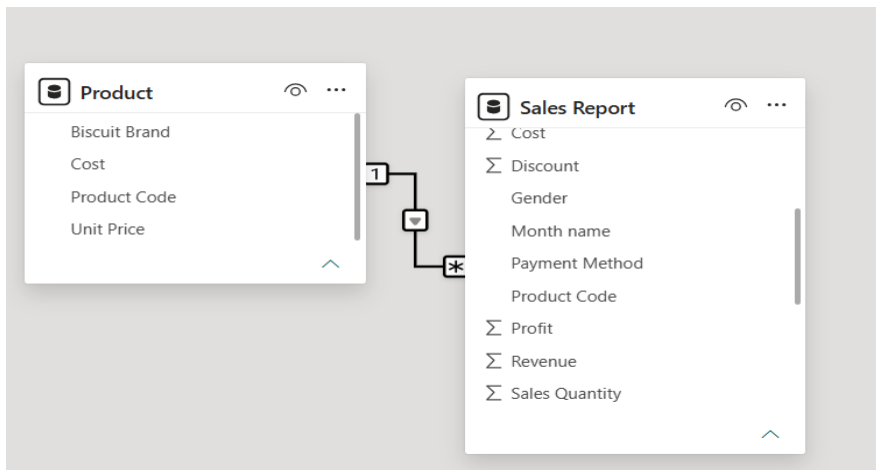
Revenue by Gender		
Gender	Revenue	Revenue
Female	₹ 225,371	4.67%
Male	₹ 4,246,882	88.03%
Other	₹ 352,255	7.30%
Revenue by Payment Method		
Payment Method	Revenue	
Cash	₹ 1,192,443	
Credit Card	₹ 1,098,472	
Debit Card	₹ 1,220,295	
Mobile Payment	₹ 1,313,298	
Revenue by Location		
Buyer Location	Revenue	
Chicago	₹ 434,322	
Dallas	₹ 496,150	
Houston	₹ 527,582	

Revenue by Top 5 Customer	
Buyer Name	Revenue
Alexander Smith	₹ 79,967
Erik Green	₹ 20,088
Justin Weeks	₹ 20,250
Mary Rodriguez	₹ 33,342
Robert Hernandez	₹ 31,852
Revenue by Monthly Trends	
Months	Revenue
Jan	₹ 849,275
Feb	₹ 697,661
Mar	₹ 736,290
Apr	₹ 573,863
May	₹ 640,058
Jun	₹ 608,248
Jul	₹ 719,113

Week Type Profit	
Week Type	Profit
Weekday	₹ 1,295,966
Weekend	₹ 532,453

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.



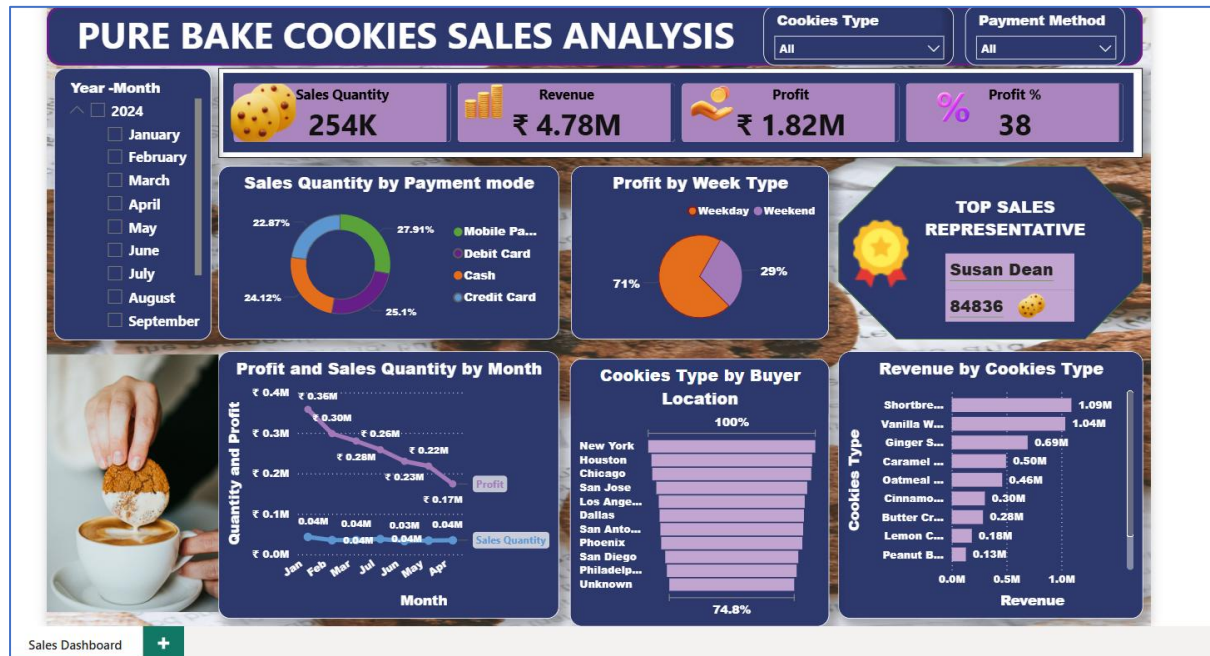
Calculated Columns Implemented key metrics, such as total sales, total revenue, profit, profit percentage, age, week day, week type, age group.

Profit % = DIVIDE(SUM('Sales Report'[Profit]),SUM('Sales Report'[Revenue]))*100													
Sales Representative	Gender	Discount	Cookies Type	Cost	UnitPrice	Age	COGS	Revenue	Profit	Week Type	Week Day Type	Age -Group	Month name
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	45	₹ 4,540	₹ 5,720.00	₹ 1,180.00	Weekday	Monday	31-45	Jan
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	36	₹ 4,280	₹ 5,393.00	₹ 1,113.00	Weekday	Tuesday	31-45	Jan
Susan Dean	Male	0.10	Shortbread	₹ 10	₹ 45	63	₹ 3,790	₹ 15,350.00	₹ 11,560.00	Weekday	Wednesday	60+	Jan
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	74	₹ 4,600	₹ 5,800.00	₹ 1,200.00	Weekday	Thursday	60+	Jan
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	19	₹ 1,750	₹ 2,205.00	₹ 455.00	Weekday	Thursday	19-30	Jan
Susan Dean	Male	0.10	Butter Crunch	₹ 10	₹ 11	19	₹ 2,940	₹ 2,930.00	30.00 -₹	Weekend	Sunday	19-30	Jan
Susan Dean	Male	0.10	Butter Crunch	₹ 10	₹ 11	47	₹ 3,590	₹ 3,554.00	36.00 -₹	Weekday	Monday	46-60	Jan
Susan Dean	Male	0.10	Shortbread	₹ 10	₹ 45	27	₹ 1,830	₹ 7,412.00	₹ 5,582.00	Weekday	Tuesday	19-30	Jan
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	47	₹ 2,330	₹ 2,900.00	₹ 60.00	Weekday	Tuesday	46-60	Jan
Susan Dean	Male	0.10	Shortbread	₹ 10	₹ 45	49	₹ 4,080	₹ 19,764.00	₹ 14,084.00	Weekday	Thursday	46-60	Jan
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	80	₹ 730	₹ 945.00	₹ 195.00	Weekday	Thursday	60+	Jan
Susan Dean	Male	0.10	Butter Crunch	₹ 10	₹ 11	59	₹ 3,700	₹ 3,663.00	37.00 -₹	Weekday	Friday	46-60	Jan
Susan Dean	Male	0.10	Butter Crunch	₹ 10	₹ 11	45	₹ 4,340	₹ 4,297.00	43.00 -₹	Weekend	Saturday	31-45	Jan
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	78	₹ 4,900	₹ 6,177.00	₹ 1,277.00	Weekday	Thursday	60+	Jan
Susan Dean	Male	0.10	Shortbread	₹ 10	₹ 45	39	₹ 4,550	₹ 18,428.00	₹ 13,878.00	Weekend	Saturday	31-45	Jan
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	37	₹ 2,510	₹ 3,163.00	₹ 653.00	Weekend	Sunday	31-45	Jan
Susan Dean	Male	0.10	Butter Crunch	₹ 10	₹ 11	24	₹ 4,470	₹ 4,425.00	45.00 -₹	Weekday	Monday	19-30	Jan
Susan Dean	Male	0.10	Butter Crunch	₹ 10	₹ 11	41	₹ 4,560	₹ 4,514.00	46.00 -₹	Weekday	Monday	31-45	Jan
Susan Dean	Male	0.10	Shortbread	₹ 10	₹ 45	61	₹ 940	₹ 3,807.00	₹ 2,867.00	Weekday	Monday	60+	Jan
Susan Dean	Male	0.10	Shortbread	₹ 10	₹ 45	19	₹ 3,400	₹ 13,770.00	₹ 10,370.00	Weekday	Wednesday	19-30	Jan
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	27	₹ 920	₹ 1,159.00	₹ 239.00	Weekday	Friday	19-30	Feb
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	60	₹ 3,200	₹ 4,032.00	₹ 832.00	Weekday	Monday	46-60	Feb
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	34	₹ 4,570	₹ 5,758.00	₹ 1,188.00	Weekday	Monday	31-45	Feb
Susan Dean	Male	0.10	Shortbread	₹ 10	₹ 45	26	₹ 4,000	₹ 16,200.00	₹ 12,200.00	Weekday	Wednesday	19-30	Feb
Susan Dean	Male	0.10	Cinnamon Swirls	₹ 10	₹ 14	21	₹ 1,640	₹ 2,066.00	₹ 426.00	Weekday	Monday	19-30	Feb
Susan Dean	Male	0.10	Butter Crunch	₹ 10	₹ 11	58	₹ 4,080	₹ 4,049.00	41.00 -₹	Weekday	Wednesday	46-60	Feb
Susan Dean	Male	0.10	Shortbread	₹ 10	₹ 45	48	₹ 1,200	₹ 4,860.00	₹ 3,660.00	Weekday	Wednesday	46-60	Feb

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:** Cards, stacked bar chart, pie charts, funnel chart, donut chart and slicers to communicate insights.
- Interactive Layout using slicers for month.
- Clear Titles for each visuals & Labels.



9. Insights & Conclusions

- **Key Findings:**
- The business generated a total **revenue of ₹4.78M** with a **profit of ₹1.82M** and an overall **profit margin of 38%**, indicating strong profitability.
- **Sales quantity reached 254K units**, showing high product demand.
- **Shortbread and Vanilla cookies** are the top revenue-generating products.
- **Weekdays contribute 71% of profit**, significantly higher than weekends (29%). **Mobile payments (27.91%) and debit cards (25.1%)** are the most preferred payment methods.
- **Susan Dean** is the top-performing sales representative.
- Cities like **New York, Houston, and Chicago** show the highest customer activity.
- Sales and profit show **fluctuations across months**, with a slight declining trend after peak months.

Analysis insights:

Descriptive:

- Total sales, profit, and quantity were analyzed across time.
- Product-wise and location-wise performance was evaluated.
- Payment method distribution and weekly trends were identified.

Insight: The company maintains strong sales volume with consistent revenue contribution from key products.

Diagnosis:

- Higher weekday profit suggests **business customers or routine buying behavior**.
- Top products (Shortbread, Vanilla) perform well due to **customer preference and brand popularity**.
- Mobile and card payments dominate due to **ease and convenience**.
- Decline in later months may be due to **seasonal demand changes or reduced promotions**.

Insight: Customer behavior, convenience, and seasonal factors strongly influence sales performance.

Predictive:

- Sales are expected to increase during peak months or promotional periods.
- High-performing products will likely continue dominating revenue.
- Digital payment usage is expected to grow further.
- If the declining trend continues, future revenue may slightly decrease.

Insight: Future growth depends on maintaining demand for top products and managing seasonal fluctuations.

Prescriptive:

- Focus on increasing production of top-selling cookies (Shortbread, Vanilla).
- Introduce offers/discounts during low-performing months to boost sales.
- Strengthen weekday promotions to maximize already high profits.
- Expand marketing in high-performing cities.
- Encourage digital payments with cashback or offers.
- Provide incentives to improve performance of other sales representatives.

Recommendation: Strategic marketing, inventory planning, and promotions can significantly improve overall performance.

10. Conclusions

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting. The Power BI dashboard effectively highlights sales performance, customer behavior, and product trends. **The business demonstrates strong profitability, but there is scope for improvement through better seasonal planning, targeted promotions, and product optimization.** By leveraging these insights, the company can enhance decision-making and achieve sustainable growth.